

UNIT 1: Physics and Human Society

Subject: Physics

Grade: Grade 9

Definition, scope, branches, related fields, and major contributors to physics in society.

Definition and Nature of Physics

The word physics comes from the Greek word phusis, meaning nature. Physics is a branch of natural science that explains the fundamental aspects of the universe: what exists, what properties things have, and what processes they undergo.

Physics helps explain how daily tools and systems work, from cars and refrigerators to radios and communication devices. It also helps us understand physical events such as motion, friction, and electrical effects.

Why Physics Matters in Human Society

Physics supports innovation and practical life by:

- explaining technology used in transport, health, and communication
- helping us understand natural events and engineering systems
- supporting learning in biology, chemistry, geology, and astronomy
- training learners to reason using concepts, principles, and laws of nature

A person who studies physics is called a physicist.

Major Branches of Physics

Physics has many branches that focus on different types of natural behavior.

Branch	Description
Mechanics	Study of motion and forces. Includes classical mechanics and quantum mechanics.
Acoustics	
Optics	Study of light, its properties, and propagation.
Thermodynamics	Study of heat, temperature, and energy transfer.
Electromagnetism	Study of electric and magnetic phenomena, including electromagnetic waves.
Nuclear Physics	
Astrophysics	Application of physics to stars, planets, galaxies, and cosmic phenomena.

Related Fields to Physics

Physics is foundational in many scientific and professional fields.

Field	How It Relates to Physics
Chemistry	Atomic and molecular behavior is explained by physical principles.
Engineering	
Geology	Earth processes such as heat flow and seismic motion use physical analysis.
Biophysics	
Geophysics	Applies physics to understand Earth structure and behavior.
Medical Physics	
	Supports imaging and treatment technologies like X-ray, MRI, and ultrasound.

Historical Issues and Key Contributors

Physics developed from classical physics to modern physics through major discoveries. This changed science, technology, and society.

Classical physics dominated from the Renaissance to the late 19th century. In the 20th century, modern physics transformed ideas about matter, energy, and the universe.

Scientist	Contribution
Galileo Galilei (1564-1642)	Pioneer of observational astronomy and experimental scientific method.
Isaac Newton (1643-1727)	
	Formulated laws of motion and universal gravitation.
Michael Faraday (1791-1867)	Advanced electromagnetism and electromagnetic induction.
James Prescott Joule (1818-1889)	
Marie Curie (1867-1934)	Pioneered research on radioactivity; discovered polonium and radium.
Albert Einstein (1879-1955)	
	Developed relativity and contributed to quantum theory.